

Cumulative Final Exam

100 Questions with Marked Breaks and Answer Bank

How to use this file: Copy the notes carefully, then cover the worked examples and redo them without looking. For practice files, work first, then check the answer bank. Detailed solutions are included for Unit 3 through Unit 8 practice as requested.

Cumulative Final Exam - 100 Questions

Directions

This final is cumulative. Round decimal answers to the nearest tenth unless an exact answer is requested. Marked breaks separate units so you can pause, check pacing, and continue. The answer bank starts after Question 100.

Unit 1: Foundations

1. Explain the difference between a line and a segment.
2. If B is between A and C , $AB = 2x + 1$, $BC = x + 8$, and $AC = 33$, find x , AB , and BC .
3. Point M is the midpoint of $\overline{AB}$. If $AM = 5x - 4$ and $MB = 3x + 10$, find x and each half.
4. Vertical angles measure $8x - 15$ and $5x + 30$. Find x and the angle measure.
5. A linear pair measures $3x + 25$ and $2x + 5$. Find x and both angles.
6. Find the distance between $(-4, 2)$ and $(8, 7)$.
7. Find the midpoint of $(-6, 9)$ and $(4, -3)$.
8. Find the slope through $(1, -2)$ and $(7, 10)$.
9. Write the equation of the line with slope 3 through $(2, -1)$.
10. Write the equation of the line parallel to $y = -4x + 6$ through $(1, 5)$.
11. Write the equation of the line perpendicular to $y = \frac{2}{3}x - 1$ through $(4, 3)$.
12. Are $y = 5x + 1$ and $-5x + y = 7$ parallel, perpendicular, or neither?

MARKED BREAK 1

End of Unit 1: Foundations. Pause, stretch, and check your pacing before continuing.

Unit 2: Transformations

13. Translate $(5, -6)$ by $(x, y) \mapsto (x - 3, y + 4)$.
14. Reflect $(-2, 7)$ over the x -axis.
15. Reflect $(-3, 8)$ over $y = x$.
16. Rotate $(-5, 2)$ 90° clockwise about the origin.
17. Rotate $(6, -9)$ 180° about the origin.
18. Reflect the line $y = 4$ over the x -axis.
19. What does a rigid transformation preserve?
20. Is dilation with scale factor 2 a rigid motion?

21. What is the smallest positive rotation that maps a square onto itself?
22. Alternate interior angles measure $7x + 4$ and $5x + 28$. Find x and the angle measure.
23. Same-side interior angles measure $4x + 12$ and $6x - 2$. Find x .
24. What can you conclude if corresponding angles made by a transversal are congruent?

MARKED BREAK 2

End of Unit 2: Transformations. Pause, stretch, and check your pacing before continuing.

Unit 3: Triangles

25. A triangle has angles $x + 20$, $2x + 10$, and $3x$. Find x and all angles.
26. An exterior angle has remote interior angles 53° and 68° . Find the exterior angle.
27. Base angles of an isosceles triangle measure $3x + 8$ and $5x - 20$. Find x , each base angle, and the vertex angle.
28. Two sides of a triangle are 5 and 12. Write the range for the third side x .
29. Classify by sides $A(0, 0)$, $B(5, 0)$, $C(2, 4)$.
30. Do sides 8, 15, 17 form a right triangle?
31. Which congruence criterion uses two sides and the included angle?
32. What does AAA prove?
33. What information is required for HL?
34. If $\triangle ABC \cong \triangle XYZ$, name the part corresponding to $\angle B$ and to AC .
35. A midsegment is parallel to a side of length 32. Find the midsegment length.
36. A median is 21 units long. How far is the centroid from the vertex?
37. Find the midpoint of the segment from $(2, 6)$ to $(8, 10)$.

MARKED BREAK 3

End of Unit 3: Triangles. Pause, stretch, and check your pacing before continuing.

Unit 4: Polygons and Quadrilaterals

38. Find the interior angle sum of a heptagon.
39. Find each interior angle of a regular nonagon.
40. Find each exterior angle of a regular 20-gon.
41. In a parallelogram, opposite angles measure $4x + 15$ and $7x - 30$. Find x and the angle.
42. Consecutive parallelogram angles measure $3x + 20$ and $5x$. Find x and both angles.

43. Diagonals of a parallelogram meet at O . If $AO = 4x - 1$ and $OC = 2x + 11$, find x and AC .
44. Rectangle diagonals measure $5x + 2$ and $7x - 16$. Find x and a diagonal length.
45. Find the area of a rhombus with diagonals 12 and 16.
46. Classify $(0, 0)$, $(4, 0)$, $(4, 6)$, $(0, 6)$ as specifically as possible.
47. Classify $(1, 0)$, $(4, 3)$, $(1, 6)$, $(-2, 3)$ as specifically as possible.
48. A quadrilateral has angles x , $2x$, $3x$, $4x$. Find x .
49. Find the perimeter of a right triangle with vertices $(0, 0)$, $(6, 0)$, $(6, 8)$.
50. Find the area of a trapezoid with bases 12 and 18 and height 5.

MARKED BREAK 4

End of Unit 4: Polygons and Quadrilaterals. Pause, stretch, and check your pacing before continuing.

Unit 5: Similarity

51. Dilate $(3, -4)$ by scale factor -2 centered at the origin.
52. Dilate $(4, 5)$ by scale factor 2 centered at $(1, 1)$.
53. A side of length 10 maps to 15. Find the scale factor from original to image.
54. Similar triangles have corresponding sides 6 and 14, and x and 21. Find x .
55. Which similarity theorem uses two pairs of congruent angles?
56. Which similarity theorem uses proportional sides around a congruent included angle?
57. Are triangles with sides 3, 4, 5 and 9, 12, 15 similar? Give scale factor.
58. In $\triangle ABC$, $DE \parallel BC$. If $AD = 8$, $DB = 12$, $AE = 10$, find EC .
59. A polygon is dilated by scale factor $\frac{5}{2}$. If the original perimeter is 28, find the image perimeter.
60. If length scale factor is 3, what is the area scale factor?
61. If area ratio is 25 : 64, what is the length ratio?
62. A 5-ft person casts an 8-ft shadow. A pole casts a 32-ft shadow. Find the pole height.

MARKED BREAK 5

End of Unit 5: Similarity. Pause, stretch, and check your pacing before continuing.

Unit 6: Right Triangles and Trigonometry

63. Find the hypotenuse of a right triangle with legs 7 and 24.
64. Find the missing leg if the hypotenuse is 17 and one leg is 8.

65. In a $30^\circ - 60^\circ - 90^\circ$ triangle, the short leg is 9. Find the long leg and hypotenuse.
66. In a $45^\circ - 45^\circ - 90^\circ$ triangle, the hypotenuse is $10\sqrt{2}$. Find each leg.
67. If opposite = 9 and hypotenuse = 15, find $\sin \theta$.
68. If opposite = 12 and adjacent = 16, find $\tan \theta$.
69. A right triangle has hypotenuse 25 and angle 40° . Find the opposite side.
70. A right triangle has adjacent side 14 and angle 55° . Find the hypotenuse.
71. Find θ if $\tan \theta = \frac{5}{12}$.
72. Rewrite $\sin 37^\circ$ as a cosine.
73. Find the area of a triangle with sides 8 and 11 enclosing 35° .
74. Use Law of Sines: $A = 50^\circ$, $a = 20$, $B = 65^\circ$. Find b .
75. Use Law of Cosines for sides 6 and 8 with included angle 120° . Find the third side.

MARKED BREAK 6

End of Unit 6: Right Triangles and Trigonometry. Pause, stretch, and check your pacing before continuing.

Unit 7: Area, Surface Area, and Volume

76. Find the area of a triangle with base 15 and height 9.
77. Find the area of a trapezoid with bases 9 and 17 and height 6.
78. Find sector area for radius 8 and central angle 45° .
79. Find arc length for radius 12 and central angle 150° .
80. Find the area of the triangle $(1, 1), (7, 1), (3, 6)$.
81. Find the volume of a cylinder with radius 5 and height 7.
82. Find the surface area of a cylinder with radius 3 and height 8.
83. Find the volume of a cone with radius 4 and height 9.
84. Find the volume of a rectangular prism with dimensions 3, 4, 10.
85. Find the surface area of a rectangular prism with dimensions 3, 4, 10.
86. Mass is 720 g and volume is 90 cm^3 . Find density.
87. Water in a cylinder of radius 4 rises by 2. Find displacement volume.
88. If length scale factor is 3, what is the volume scale factor?

MARKED BREAK 7

End of Unit 7: Area, Surface Area, and Volume. Pause, stretch, and check your pacing before continuing.

Unit 8: Circles

89. Write the equation of a circle with center $(-2, 5)$ and radius 6.
90. Find the center and radius of $(x - 7)^2 + (y + 3)^2 = 64$.
91. For a circle centered at $(0, 0)$ with radius 13, classify point $(5, 12)$.
92. A circle has center $(1, 2)$ and passes through $(4, 6)$. Write its equation.
93. Find the tangent slope at $(5, 12)$ on a circle centered at $(0, 0)$.
94. An inscribed angle intercepts arc 110° . Find the angle.
95. A tangent-chord angle intercepts arc 130° . Find the angle.
96. An exterior angle intercepts arcs 200° and 80° . Find the angle.
97. Intersecting chords satisfy $5 \cdot 9 = x \cdot 15$. Find x .
98. A tangent and secant share an external point. The secant has external part 8 and internal part 10. Find the tangent length.
99. A cyclic quadrilateral has an angle of 112° . Find its opposite angle.
100. A regular hexagon is inscribed in a circle of radius 14. Find its side length.

Final Exam Answer Bank

1. A line extends forever in both directions; a segment has endpoints and finite length.
2. $x = 8$, $AB = 17$, $BC = 16$.
3. $x = 7$; each half is 31.
4. $x = 15$; angle 105° .
5. $x = 30$; angles 115° and 65° .
6. 13.
7. $(-1, 3)$.
8. 2.
9. $y = 3x - 7$.
10. $y = -4x + 9$.
11. $y = -\frac{3}{2}x + 9$.
12. Parallel.
13. $(2, -2)$.
14. $(-2, -7)$.
15. $(8, -3)$.
16. $(2, 5)$.
17. $(-6, 9)$.
18. $y = -4$.
19. Lengths and angle measures, so shape and size.
20. No.
21. 90° .
22. $x = 12$; angle 88° .
23. $x = 17$.
24. The lines are parallel.
25. $x = 25$; angles 45° , 60° , 75° .
26. 121° .
27. $x = 14$; base angles 50° ; vertex 80° .
28. $7 < x < 17$.
29. Isosceles.
30. Yes.
31. SAS.
32. Similarity, not congruence.
33. Right triangles with congruent hypotenuse and one congruent leg.
34. $\angle Y$; XZ .
35. 16.
36. 14.
37. $(5, 8)$.
38. 900° .
39. 140° .
40. 18° .
41. $x = 15$; angle 75° .
42. $x = 20$; angles 80° , 100° .
43. $x = 6$; $AC = 46$.
44. $x = 9$; diagonal 47.
45. 96.
46. Rectangle, not square.
47. Square.
48. 36.
49. 20.
50. 75.
51. $(-6, 8)$.
52. $(7, 9)$.
53. $\frac{3}{2}$.
54. 9.
55. AA.
56. SAS similarity.
57. Yes; scale factor 3.
58. 15.
59. 70.

60. 9.
61. $5 : 8$.
62. 20 ft.
63. 25.
64. 15.
65. Long leg $9\sqrt{3}$, hypotenuse 18.
66. 10.
67. $\frac{3}{5}$.
68. $\frac{3}{4}$.
69. 16.1.
70. 24.4.
71. 22.6° .
72. $\cos 53^\circ$.
73. $44 \sin 35^\circ \approx 25.2$.
74. 23.7.
75. $2\sqrt{37} \approx 12.2$.
76. 67.5.
77. 78.
78. 8π .
79. 10π .
80. 15.
81. 175π .
82. 66π .
83. 48π .
84. 120.
85. 164.
86. 8 g/cm^3 .
87. 32π .
88. 27.
89. $(x + 2)^2 + (y - 5)^2 = 36$.
90. Center $(7, -3)$, radius 8.
91. On the circle.
92. $(x - 1)^2 + (y - 2)^2 = 25$.
93. $-\frac{5}{12}$.
94. 55° .
95. 65° .
96. 60° .
97. 3.
98. 12.
99. 68° .
100. 14.